Lecture 28.06: Information Gain & Split Selection

📌 Topic Overview

How does a decision tree algorithm decide which feature gives the best split? The answer is **Information Gain (IG)**. In this lecture, we study the reduction-in-entropy formula and work through the exact numerical example taught on the blackboard.

🎯 What is Information Gain?

> **Information Gain measures the reduction in entropy (uncertainty) achieved after splitting a dataset based on a given feature.**

The intuition is simple:

$$extInformationGain = extEntropyofParent - extWeightedEntropyofChild$$

Mathematical Formula:

> IG(S, A) = H(S) - \sum_{v \in ext{Values}(A)} rac{|S_v|}{|S|} H(S_v)

Where:

- S = Parent dataset at the current node.
- $H(S)$ = Entropy of the parent node.
- A = Feature we are evaluating.
- v = A specific value or branch of feature A.
- S_v = Subset of samples where feature $A = v$.
- $rac|S_v||S|$ = Proportion of samples going down branch v (weight).

🧮 Blackboard Numerical Walkthrough

1. Parent Node Entropy

Consider the parent dataset with 6 samples: $[2y, 4n]$ (2 Yes, 4 No).

$$H(S) = - \frac{2}{6}\log_2\left(\frac{2}{6}\right) - \frac{4}{6}\log_2\left(\frac{4}{6}\right) \approx 0.918$$

---

2. Candidate Feature 1: `Gender`

Splitting on `Gender` (Male vs Female):

- **Male branch (S_M)**: Contains 3 samples: $[2y, 1n]$.

$$H(S_M) = - \frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right) \approx 0.918$$

- **Female branch (S_F)**: Contains 3 samples: $[0y, 3n]$ (all No $\rightarrow$ Pure).

$$H(S_F) = 0.0$$

Now compute weighted child entropy:

$$\text{Weighted Child Entropy} = \frac{3}{6} \cdot H(S_M) + \frac{3}{6} \cdot H(S_F) = \frac{1}{2}(0.918) + \frac{1}{2}(0) = 0.459$$

Now compute Information Gain for Gender:

$$IG(extGender) = H(S) - 0.459 = 0.918 - 0.459 = 0.459$$

---

3. Candidate Feature 2: `Employed`

The instructor notes on the board:

> $$IG(extEmployed) = 0.800$$

---

4. Split Selection Decision

We compare the Information Gain of all candidate features:

- $IG(extGender) = 0.459$
- $IG(extEmployed) = 0.800$

Since $0.800 > 0.459$, the tree selects `Employed` as the split feature at this node!

💻 Python Implementation

```
import numpy as np

def entropy(y, n):
total = y + n
if total == 0 or y == 0 or n == 0:
return 0.0
py = y / total
pn = n / total
return -(py * np.log2(py) + pn * np.log2(pn))

def info_gain(parent_y, parent_n, left_y, left_n, right_y, right_n):
total_p = parent_y + parent_n
total_l = left_y + left_n
total_r = right_y + right_n

h_parent = entropy(parent_y, parent_n)
h_left = entropy(left_y, left_n)
h_right = entropy(right_y, right_n)

weighted_child_entropy = (total_l / total_p) * h_left + (total_r / total_p) * h_ri
return h_parent - weighted_child_entropy

# Feature 1: Gender split
ig_gender = info_gain(2, 4, 2, 1, 0, 3)
print(f"Parent Entropy: {entropy(2, 4):.3f}")
print(f"IG (Gender):   {ig_gender:.3f}")
```

🔑 Key Takeaways

1. Information Gain represents the reduction in entropy caused by a split.
2. The feature with the **highest Information Gain** is chosen as the split node.
3. When using Gini Impurity, the equivalent metric is **Gini Gain**.